MOHAMMAD REHAN



Founder & Lead Systems Architect @ Rehan Quant Lab™

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EXECUTIVE SUMMARY

Founder & Lead Systems Architect at Rehan Quant Lab with 4+ years of hands-on experience bridging mission-critical commercial operations and fault-tolerant distributed systems. Creator of the Deterministic State Kernel (bare-metal Rust engine with >240,000 writes/sec, 604ns reads, hardware SHA-256 WAL, and MCA Rule 11(g) compliance), AegisDB (2.54M ticks/sec @ 0.39µs via LMAX Disruptor cacheline alignment, 21.4M AVX2 SIMD invariant ops/sec @ 46.69ns), and the Antifragile Triad (Aegis Pre-Audit Chaos Injection Phoenix <500ms Resurrection with Rs. 0.00 capital loss). Architect of pure native Windows .EXE air-gapped sovereign trading suites deployed on AWS Mumbai.

CORE COMPETENCIES & ARCHITECTURE STACK

- **Deterministic State Kernel (Rust Bare-Metal Engine)**: 100% on-premises, air-gapped state invariants engine compiled into native C-ABI dynamic library (`dse_core.dll`) via Fat LTO. Delivers >240k writes/sec (4.69µs commit), 604ns cached reads (<1µs), hardware SHA-256 Write-Ahead Log (WAL), and cold crash replay of 25,000 frames in 76.99ms with zero cloud leaks.
- **The 4 Invariant Principles**: Non-Newtonian Liquid Risk Armor (continuous volatility damping & dynamic spread clamping), Hardware SHA-256 Trade WAL, Invariant Risk Enforcement, and Air-Gapped Sovereign Execution (zero cloud telemetry, Rs. 0 recurring SaaS bills).
- **The Antifragile Triad (The Holy Trinity)**: Aegis Shield pre-audit contract verification, Chaos Gauntlet adversarial fault injection (8 market disaster simulations), and Phoenix Autonomous Recovery (<500ms self-healing with Rs. 0.00 capital loss and genomic mutation).
- **AegisDB In-Memory Engine**: LMAX Disruptor 64-byte cacheline-padded ring buffer (2.54M ticks/sec, 0.39µs latency, zero false sharing across CPU cores), bare-metal RDTSC JIT x86_64 cycle counter (38.04 cycles/op), memory-mapped LSM WAL (3.56M syncs/sec, 3.02ms kill -9 hot recovery).
- **AVX2 SIMD Vectorized Invariant Engine**: High-throughput vectorized financial boundary evaluation across 10,000,000 operations (21.4M ops/sec, 46.69ns latency), enforcing 35% drawdown clamps, 30% untouchable cash reserve, and 0.000000000000% precision drift.
- **Pure Native Windows Executables**: 100% pure native PE binaries (`.exe`), strictly zero shell scripts (`.bat`), fully air-gapped with embedded console pauses for enterprise deployment trust.
- **FinTech & Algorithmic Trading**: Systematic options execution (NSE NIFTY/BANKNIFTY), exchange-native Stop-Loss Limit orders with 1.5% protection buffer, Spot/15m VWAP structural stop-loss guards, sub-50ms broker API roundtrip, Kotak Neo v3 orchestrator.
- **Cloud Infrastructure & Security**: AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Zero-Trust Network Access (ZTNA) via Private WireGuard/Tailscale Mesh, Linux systemd real-time priority tuning (Nice=-10).

KEY PRODUCTION SYSTEMS & ARCHITECTURES

Deterministic State Kernel -- Sub-Microsecond Rust State & Risk Engine

- Engineered a commercial bare-metal Rust state invariants engine compiled with Fat LTO (`opt-level = 3`, `codegen-units = 1`, `panic = "abort"`) producing a compact 1.06 MB C-ABI dynamic library (`dse_core.dll`) and 475 KB CLI (`dse-kernel-cli.exe`).
- Benchmarked write throughput exceeding 240,000 writes/sec (4.69µs commit latency) and read throughput exceeding 1,570,000 reads/sec (604ns cached latency).
- Implemented hardware SHA-256 Write-Ahead Logging with cold crash replay of 25,000 state frames in 76.99ms with zero data drift, satisfying MCA Rule 11(g) and SEBI quant audit trail requirements.
- Developed universal Python `ctypes` FFI bridge achieving >124,000 writes/sec directly from high-level Python runtimes without GIL contention.

Indian Quant Options Sniper -- Institutional Derivatives Execution Engine

- Engineered a systematic algorithmic options execution engine deployed on AWS Mumbai (ap-south-1) with connection-pooled WebSocket market streaming and sub-50ms broker order dispatching.
- Integrated Dynamic Volatility-Adaptive Sizing & Spread Clamping: Automated entry lockout on spread widening (>= Rs. 1.50) and position compression during volatility expansion.
- Deployed 7-Stage Multi-Factor Execution Tribunal with Directional Toxicity Intelligence (VPIN/CVD) distinguishing aggressive momentum sweeps from hostile seller dumps.
- Submerged `dse_core.dll` native Rust engine and AegisDB ring buffer for sub-microsecond (<0.39µs) tick ingestion and active trade state caching.
- Anchored dynamic stop-loss to NIFTY Spot 15m VWAP structure rather than option premium noise, backed by a 35% emergency disaster ceiling and a 30% capital reserve lock.

AegisDB -- Distributed Antifragile In-Memory Database (Open-Source MIT)

- Architected an ultra-low-latency in-memory state engine and circular tick ring buffer with 64-byte cacheline padding achieving 2,535,916 ticks/sec at 0.39µs latency with zero false sharing across CPU cores.
- Implemented bare-metal x86_64 machine code JIT timing (RDTSC) reading physical CPU cycles directly in 38.04 cycles per operation.
- Evaluated 10,000,000 extreme market operations in AVX2 SIMD vector chunks at 21,417,114 ops/sec (46.69ns per invariant check) with audited 0.000000000000% mathematical precision drift.
- Built a zero-syscall memory-mapped (mmap) LSM append-only Write-Ahead Log (WAL) achieving 3,559,925 syncs/sec (0.28µs append) and 3.02ms kill -9 hot replay recovery with Rs. 0.00 / 0 bytes data loss.

Aegis Crypto Radar -- 24/7 Asynchronous Microstructure Surveillance Engine

- Engineered 24/7 asynchronous global crypto surveillance engine analyzing orderbook depth (WOBI), cross-exchange basis divergence (Binance vs Coinbase), whale aggression (CVD), and perpetual funding squeezes.
- Submerged `dse_core.dll` bare-metal Rust engine for real-time sub-microsecond telemetry caching and SHA-256 WAL state recording.
- Fortified with 6-disaster Chaos Gauntlet and Phoenix Healer autonomous recovery watchdog (<500ms MTTR). 19/19 Pytest suites passed with 100% test coverage.

Footwear Retail POS & High-Volume Inventory Ledger

- Built a commercial Point of Sale (POS) and inventory ledger managing retail store operations, barcode scanning, thermal printing, and GST compliance.
- Features multi-counter SQLite WAL concurrency protection, inventory reorder forecasting, and real-time sales margin dashboards.
- Engineered automated staff attendance logging and stock replenishment forecasting, reducing inventory holding costs by 30%.

PROFESSIONAL EXPERIENCE

Founder & Lead Systems Architect

Rehan Quant Lab -- Patna, Bihar, India (Remote Global) *(January 2023 - Present)*

- Direct independent systems research and software development for ultra-low-latency in-memory databases, autonomous trading bots, and deterministic quantitative trading infrastructure.
- Architected Deterministic State Kernel in Rust (>240k writes/sec, 604ns reads, SHA-256 WAL) commercialized globally on Gumroad ($249 / Rs. 19,999).
- Engineered AegisDB, achieving 2.62M reads/sec (0.38µs) and sub-15ms WAL crash resurrection with zero-heap garbage collection jitter.
- Deployed Indian Quant Options Sniper 24/7 on AWS EC2 Mumbai (ap-south-1) with connection-pooled WebSocket market feeds.
- Built the 3-AZ Fault-Tolerant Quorum Architecture implementing Nassim Taleb's Golden Rule of Resilience with continuous evolutionary parameter adaptation.

Store & Operations Manager

Ladies Zone Exclusive Showroom -- Patna, Bihar, India *(March 2021 - Present)*

- Direct day-to-day operations, inventory replenishment, and customer relationship management for a high-volume retail fashion showroom in Patna, Bihar.
- Architected and deployed in-house biometric attendance audit ledger, retail POS billing software, and custom business workflow automations.
- Bridged physical business requirements with automated software engineering to eliminate operational overhead and inventory discrepancies.
- Managed vendor relations, staff scheduling, commercial margin analysis, and customer dispute resolution with 100% operational continuity.
- Spearheaded inventory digital transformation, cutting dead-stock capital allocation by 25% through real-time stock turnover telemetry.

EDUCATION & ACADEMIC CREDENTIALS

Bachelor of Computer Applications (BCA)

National Business College -- Patna, Bihar, India *(Graduated January 2025)*

- Core coursework: Object-Oriented Software Design, Database Management Systems, Data Structures, Web Technologies, and Network Security.
- Capstone Project: Enterprise Retail Management System with Real-Time Inventory Tracking and POS Integration.

TECHNICAL ENVIRONMENT & PRODUCTION SAFEGUARDS

- **Languages & Core Systems**: Rust (Bare-Metal C-ABI FFI, Fat LTO), C (AVX2 SIMD), Python 3.14/3.12, AsyncIO, FastAPI, Streamlit, Modern JavaScript, Node.js.
- **Databases & Protocols**: AegisDB (In-Memory Ring Buffers), Deterministic State Kernel (Rust WAL), SQLite (WAL micro-journaling), RESTful APIs, WebSocket, WireGuard/Tailscale VPN.
- **Architectural Principles**: The 4 Invariant Principles, Antifragile Triad (Aegis-Chaos-Phoenix), Zero Capital Leakage, Atomic Single-Route Execution, Pure Native .EXE (Zero .bat).
- **DevOps & Infrastructure**: AWS Mumbai (ap-south-1), Cloudflare Pages Edge, Linux systemd (Nice=-10), Git, 250+ Unit & Chaos Invariant Tests (100% Passed).